

THE EFFECT OF PROGESTERONE AND PROGESTOGENS ON THE FOETUS

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In order to appreciate the effect of progesterone and progestogens on the foetus it is essential that you are able to fully understand the vital differences between naturally occurring progesterone and synthetic progestogens. The standard assay test for progestational activity is the Clauberg test, in which immature female rabbits, primed with oestrogen, show the glandular changes in the endometrium characteristic of progesterone. But this test was devised in the forties, when the other properties of progesterone were unknown, in particular there was complete ignorance of the widespread distribution of progesterone receptors throughout the body, and no knowledge of its immunosuppressive or thermogenic properties. It was recognized that progesterone was inactive when administered orally, and to overcome this the synthetic progestogens were developed from testosterone. Progestogens gave positive Clauberg tests and could be administered orally. They mimicked the action of progesterone on the endometrium but NOT the other properties, indeed administration of progestogens lowers the blood progesterone level (Johansson, 1971), whereas the administration of progesterone raises the blood progesterone level (Nillius and Johansson, 1971). Progestogens are adrogenic and anabolic, and some are oestrogenic.

Progesterone receptors transport molecules of progesterone through the cell wall and cytoplasm to the cell nucleus to modify gene expression. Progesterone receptors are present throughout the body with high concentrations in the liver, uterus and mid-brain. Orally administered progesterone, which passes through the portal system to the liver, is metabolized in the liver before reaching the foetal circulation. These progesterone receptors are highly specific and do not metabolize progestogens, or even the isomers of progesterone (Yen and Jaffe, 1978); so progestogens, administered orally, can pass via the liver into the systemic circulation. Considerable metabolism of progesterone occurs in the foetal tissues, including the foetal brain, and Michan and Zander (1979) have shown that in the eleventh week of foetal life a significant amount of *de novo* synthesis of progesterone is occurring although progesterone is metabolized differently in the foetal brain, there being no 5 α reduced sulphates isolated.

The effect of progesterone and progestogens on the foetus will be considered separately.

For the past 30 years I have been using progesterone in pregnancy, first by injection and in later years by suppositories, for the relief of such pregnancy symptoms as headaches, vomiting, depression and lethargy. In a survey of 633 pregnant women it was found that 25% of women with pregnancy symptoms between the 16 and 28th week of pregnancy later developed pre-eclampsia, compared with only 10% of women who were free from these symptoms during the single interval in the middle trimester (Dalton, 1960). Two controlled trials have shown that, by using either injections (Dalton, 1962) or suppositories (Dalton, 1977), the incidence of pre-eclampsia can be reduced to 3%. It is now appreciated that progesterone has immunosuppressive properties, so it is of interest that present thinking suggests that pre-eclampsia is an immunological disease. Pre-eclampsia has been associated with diminished intelligence in the child of that pregnancy (Barker and Edwards, 1967) so careful follow-up studies were performed on children of mothers who received progesterone antenatally, (these will be referred to as "progesterone children"). Fortyfour progesterone children were matched with two control groups. A "normal control", consisting of the nextborn child in the labour ward register whose mother had a normal pregnancy and labour, and a "toxaemic control" whose mother had pre-eclampsia and received no progesterone. An educational follow-up done on these children 9-10 years later by schoolteachers on a blind basis, revealed significantly more "above average" grades among progesterone children compared with controls. These results were significant for all academic subjects, arithmetic, English and verbal reasoning but not for physical education or craftwork. Furthermore there was a positive correlation between "high dose" and "early treatment before the 16th week" of pregnancy on the one hand and above average achievement on the other hand (Dalton, 1968).

Further follow-up showed that at 16 years these progesterone children had more "O" levels and at 18 years more "A" levels than the controls. Furthermore 32% of the 34 progesterone children entered a British university compared with 6% of control children (Dalton, 1976). Indeed, 6% of all children in the Borough of Haringey, in which the hospital is situated and 6% of children from the Inner London Education Authority entered university. Again the progesterone

children's performance was related to dose, administration before the 16th week and duration over 8 weeks.

Interest in these findings was aroused in Stanford University who sent two psychologists, Zussman, Zussman and Dalton (1975) over to England to do full psychological testing of the progesterone children. Their findings confirmed high numerical ability associated with both high dose ($P = 0.003$) and longer duration ($P = 0.023$). On the psychological and social development they found no differences; especially important was their findings of similarity of the groups in the masculinity-femininity scores on the Bem Sex role inventory. Although not statistically significant the progesterone children were noted to be less active in sport and play and more concerned with hair and clothes, suggesting that the ante-natal effect was of feminization rather than the masculinization which is recognized to follow progestogen administration.

Lynch, Mychalkiw and Hutt (1978) from Keele University also expressed interest in the findings and were given assistance enabling them to follow up the progesterone and control children, whose mothers were included in the two controlled trials into the use of progesterone in the prophylaxis of pre-eclampsia, carried out at Chase Farm Hospital, Middlesex (Dalton, 1962, 1977). In these trials no progesterone was administered before the 16th week. Women with pregnancy symptoms during the middle trimester were allocated at random to the progesterone or control groups. Progesterone injections had been used in the trial of children tested at 16 years, and progesterone suppositories in children tested at two years. Less than 20% of children were traced, but they found no differences between the progesterone and control groups in their mental or social development, or their spatial, perceptual or personality tests. This suggests that the beneficial effect of progesterone is limited to early administration, as none of these children received progesterone before the 16th week, so probably, as with rubella and thalidamide, there may be a period of maximal sensitivity to progesterone during the first 16 weeks when foetal brain development is occurring.

It should be mentioned that in all these studies there have been no instances of foetal abnormalities, nor in my 30 years of using progesterone in pregnancy have I seen masculinization in a girl whose mother received progesterone during pregnancy.

Now to turn to progestogens. The first reported side effects of progestogens were masculinization of the external genitalia, with normal internal gonads (Wilkins, 1960). This is usually partial, amounting to enlargement of the clitoris and labia, and only rarely is there complete fusion with the formation of a penis. There is one recorded case in the literature, and another known to me, of true hermaphroditism following administration of medroxyprogesterone, (Jones, Ferguson-Smith and Heller, 1965). On the other hand, abnormalities known to follow adminis-

tration of oestrogen-progestogen preparations are more varied and serious including spina bifida (Gal, Kirman and Stern, 1967); multiple deformities (Nora and Nora, 1973); oesophagel anomalies (Oakley, Flint and Falek, 1973); heart defects (Levy, Cohen and Fraser 1973); and limb reduction deformities (Janerich, Piper and Glebatis, 1973). An interesting observation by Janerich and colleagues was that the limb reduction deformities among those whose mothers had used oestrogen-progestogen combinations was limited to males. A recent Japanese study (Matsunaga and Shida, 1979) examined 667 fetuses obtained at termination of which 130 had been subjected to oestrogen-progestogen preparations for threatened abortion. These workers concluded that major malformations recognizable at the embryonic stage were not related to the hormone preparations.

On the behavioural aspect, Ehrhardt and Money (1967) noted a raised IQ in 10 girls masculinized by progestogen in early pregnancy, and noted their tendency to be regarded as tomboys. However, Ehrhardt, Grisanti and Meyer-Bahlburg (1977), in studies of children who had received medroxyprogesterone antenatally, found no evidence of tomboyism among 15 girls who in fact had a preference for feminine clothing styles. Among the 10 boys they noted less intense expenditure of energy, and 2 boys were teased for effeminacy, of whom one was of short stature and with a congenital heart and malformed feet (Meyer-Bahlburg, Grisanti and Ehrhardt, 1977).

Thus, progesterone appears to improve intellectual potential if administered before the 16th week, in high dose and for at least 8 weeks, and allows normal psychological social development, without causing anatomical abnormalities, nor masculinization or feminization. In contrast progestogens, especially those of the 19 nor-testosterone group, when used in combination with oestrogens, may cause anatomical abnormalities and also psychological masculinization.

There is also a condition of adrenogenital syndrome, in which owing to an enzyme deficiency, the adrenal cortex fails to synthesize cortisone and instead there is an accumulation of androgens. The girls are born with masculinization of the external genitalia and so usually diagnosed at birth. The boys appear normal at birth, although may be recognized due to excess salt loss soon after birth, or in early childhood they show signs of premature puberty. Treatment is lifelong cortisone. It is due to a genetic trait which is carried by both parents. Money and Lewis (1965) have noted a raised IQ in those suffering from adrenogenital syndrome and also in siblings of those with adrenogenital syndrome.

The part played in pregnancy by progesterone is species specific. In rabbits, removal of the ovaries during pregnancy results in an abortion, although the pregnancy can be maintained by progesterone administration. In monkeys and humans the ovary may be removed after the first trimester of pregnancy without interruption of the pregnancy, which is maintained by

the progesterone output of the placenta. Goy (1968) at the Oregon Primate Research Centre demonstrated that pregnant rhesus monkeys which were given testosterone or progestogens resulted in the female offspring showing masculinization of the external genitalia and increased tendency to masculine play behaviour, enjoying the rough and tumble and chasing play during the prepubertal years, although their sex hormone levels were similar to normal female monkeys. However, at puberty they oophorectomized these androgenized females and treated them first with oestrogen, which increased their aggressive behaviour, and then with testosterone which further increased the aggression in comparison with normal females. Thus, if androgens are given at a critical time in foetal development the exposed animal will exhibit more masculine behaviour when stimulated by androgens in adulthood.

Money (1971) reminds us that for every known instance of a masculinized daughter there are probably hundreds that were unaffected, as well as hundreds of apparently normal sons, who cannot now be traced for testing.

We are now seeing the second generation of pill takers. Recently, I have seen a few girls between 17 and 20 years, who have behaved normally during their prepubertal years suddenly develop behaviour problems, with violence, confusion and psychosis at adolescence, and who on full examination are noted to be hirsute and have raised testosterone and/or androstendione and low sex hormone binding globulin levels. I say full examination, as often the girls are wearing jeans to cover their hairy legs, have waxed their faces and had electrolysis on their breast hairs.

And now to turn from scientific to anecdotal medicine. I have 9 such girls under my care at present in which the mother revealed, on questioning, that she had received tablets during pregnancy, which she believed to have been hormones for the confirmation of pregnancy, or treatment of hyperemesis, threatened abortion, ovarian cyst or pre-eclampsia. These girls have responded well to cyproterone acetate, both in relief of hirsutism and normalising behaviour. Could it be that the hormone tablets administered antenatally sensitized the girls, so that when their own hormonal output started at puberty, or when they took the pill it resulted in increasing their aggressive behaviour and their hirsutism? How much can you rely on the memory of a woman 20 years later?

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